ECON 182: SECTIONS

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Week 3, Winter 2025

PLAN FOR TODAY

1. Labor market facts
2. Matching function

QUESTION 1

Suppose a country has 200 million people, of whom 50 million are retired, 30 million are above 16 and at university, 70 million are below 16, 10 million are in the armed forces, 30 million are working, and 10 million are not working but do not want a job. What is the labor-force participation rate?

(A) 10%

(B) 25%

(C) 30%

(D) 50%

(E) 60%

(F) 70%

(G) None of the above

$$\frac{30}{200 - 70 - 10} \times 100$$

$$= \frac{30}{120} \times 100$$

$$= 25\%$$

QUESTION 2

Monthly job finding rate = 0.5

In the United States, the average amount of time people spend unemployed is:

(A) Between zero and one month

(B) About two months

(C) About four months

(D) About six months

(E) Greater than twelve months

(F) None of the above

Avg. time people spend unemployed $= \frac{1}{0.5}$

$= 2$ months

QUESTION 3

Monthly job separation rate $\simeq 3\%$

In the United States, the average amount of time people keep a given job is:

(A) Less than one month

(B) Between two and four months

(C) About one year

(D) Between two and four years

(E) More than five years

(F) None of the above

Avg. duration of job $= \frac{1}{3\%}$

$= \frac{100}{3}$

$\simeq 33$ months

QUESTION 4

Over the US business cycle, how do the unemployment and vacancy rates comove?

(A) The unemployment and vacancy rates are acyclical.

(B) Both unemployment and vacancy rates are procyclical.

(C) Both unemployment and vacancy rates are countercyclical.

(D) The vacancy rate is countercyclical while the unemployment rate is procyclical.

(E) The vacancy rate is procyclical while the unemployment rate is countercyclical.

(F) None of the above.

QUESTION 5

$m(\lambda U, \lambda V) = \lambda m(U, V)$ $\lambda > 0$

$m(\lambda U, \lambda V) = \lambda m(U, V)$

Which of these matching functions does not have constant returns to scale?

(A) $m(U, V) = a \times U + b \times V$

(B) $m(U, V) = U^a \times V^{1-a}$

(C) $m(U, V) = [b \times U^a + (1 - b) \times V^a]^{1/a}$

(D) $m(U, V) = U \times V$

(E) $m(U, V) = \sqrt{U} \times \sqrt{V}$

(F) None of the above

(A) $m(u, v) = au + bv$

$$m(\lambda u, \lambda v) = a(\lambda u) + b(\lambda v)$$
$$= \lambda \underbrace{[au + bv]}_{m(u,v)}$$
$$= \lambda\, m(u, v)$$

CRS

(B) $m(u, v) = u^a v^{1-a}$

$$m(\lambda u, \lambda v) = (\lambda u)^a (\lambda v)^{1-a}$$
$$= \lambda^{a+1-a} \underbrace{u^a v^{1-a}}_{m(u,v)}$$
$$= \lambda\, m(u, v)$$

CRS

(C) $m(u, v) = \left[b u^a + (1-b) v^a \right]^{1/a}$

$$m(\lambda u, \lambda v) = \left[b(\lambda u)^a + (1-b)(\lambda v)^a \right]^{1/a}$$

$$= \left[b \lambda^a u^a + (1-b) \lambda^a v^a \right]^{1/a}$$

$$= \underbrace{\lambda^{a \cdot 1/a}}_{\lambda} \left[\underbrace{b u^a + (1-b) v^a}_{m(u,v)} \right]^{1/a}$$

CRS

(D) $m(u, v) = u \times v$

$$m(\lambda u, \lambda v) = \lambda u \times \lambda v$$

NOT

$$= \lambda^2 m(u, v)$$

CRS

QUESTION 6

$$q = \frac{m(U, V)}{V}$$

A Cobb-Douglas matching function gives the flow of new worker-firm matches created when they are U unemployment workers and V vacancies: $m = \omega \times U^{\eta} \times V^{1-\eta}$. We define labor market tightness as $\theta = V/U$. What is the expression for the rate q at which a vacancy is filled?

(A) $q(\theta) = \omega \times \theta^{\eta}$

(B) $q(\theta) = \omega \times \theta^{1-\eta}$

(C) $q(\theta) = \omega \times \theta^{-\eta}$

(D) $q(\theta) = \omega \times \eta^{\theta}$

(E) $q(\theta) = \theta^{-\eta}$

(F) None of the above

$$q = \frac{\omega \times U^{\eta} \times V^{1-\eta}}{V}$$

$$= \omega \times U^{\eta} \times V^{1-\eta-1}$$

$$= \omega \times U^{\eta} \times V^{-\eta}$$

$$= \omega \times \left(\frac{V}{U}\right)^{-\eta}$$

$$= \omega \times \theta^{-\eta}$$

QUESTION 7

$$f = \frac{m(U, V)}{U}$$

$$\frac{1}{x} = x^{-1}$$

A Cobb-Douglas matching function gives the flow of new worker-firm matches created when they are U unemployment workers and V vacancies: $m = \omega \times U^{\eta} \times V^{1-\eta}$. We define labor market tightness as $\theta = V/U$. What is the expression for the rate f at which a worker finds a job?

(A) $f(\theta) = \omega \times \theta^{\eta}$

(B) $f(\theta) = \omega \times \theta^{1-\eta}$

(C) $f(\theta) = \omega \times \theta^{-\eta}$

(D) $f(\theta) = \omega \times \eta^{\theta}$

(E) $f(\theta) = \omega \times \theta^{1+\eta}$

(F) None of the above

$$f = \frac{w \times U^{\eta} \times V^{1-\eta}}{U}$$

$$= w \times U^{\eta-1} \times V^{1-\eta}$$

$$= w \times \frac{V^{1-\eta}}{U^{1-\eta}}$$

$$= w \times \theta^{1-\eta}$$

QUESTION 8

$f = \frac{m}{U}$, $q = \frac{m}{V}$, $\theta = \frac{V}{U}$

For any matching function, what is a key relationship between the job-finding rate f, vacancy-filling rate q, and labor market tightness θ?

(A) $f + q = \theta$

(B) $f \times q = \theta$

(C) $f/q = \theta$

(D) $f - q = \theta$

(E) $q/f = \theta$

(F) None of the above

$$\frac{f}{q} = \frac{\cancel{m}}{U} \times \frac{V}{\cancel{m}} = \frac{V}{U} = \theta$$

QUESTION 9

$$f = \frac{m(U,V)}{U}$$

We define labor market tightness as $\theta = V/U$. What is the rate at which unemployed workers find a job when the matching function is $m(U,V) = (U^{-\gamma} + V^{-\gamma})^{-\frac{1}{\gamma}}$ with $\gamma > 0$?

(A) $f(\theta) = (1 + \theta^{\gamma})^{-\frac{1}{\gamma}}$

(B) $f(\theta) = (1 + \theta^{-\gamma})^{\gamma}$

(C) $f(\theta) = (1 + \theta^{-\gamma})^{-\frac{1}{\gamma}}$

(D) $f(\theta) = (1 + \theta^{\gamma})^{\gamma}$

(E) $f(\theta) = (1 - \theta^{\gamma})^{-\frac{1}{\gamma}}$

(F) None of the above

$$f = \frac{(U^{-\gamma} + V^{-\gamma})^{-\frac{1}{\gamma}}}{U}$$

$$= \frac{(U^{-\gamma} + V^{-\gamma})^{-\frac{1}{\gamma}}}{U^{-\gamma \cdot -\frac{1}{\gamma}}}$$

$$= \left(\frac{U^{-\gamma} + V^{-\gamma}}{U^{-\gamma}}\right)^{-\frac{1}{\gamma}}$$

$$= \left[1 + \underbrace{\left(\frac{V}{U}\right)}_{\theta}^{-\gamma}\right]^{-\frac{1}{\gamma}}$$

QUESTION 10

$$q = \frac{m(U,V)}{V}$$

We define labor market tightness as $\theta = V/U$. What is the rate at which vacancies are filled when the the matching function is $m(U,V) = (U^{-\gamma} + V^{-\gamma})^{-\frac{1}{\gamma}}$ with $\gamma > 0$?

(A) $q(\theta) = (1 + \theta^{-\gamma})^{-\frac{1}{\gamma}}$

(B) $q(\theta) = (1 + \theta^{\gamma})^{\gamma}$

(C) $q(\theta) = (1 + \theta^{\gamma})^{-\frac{1}{\gamma}}$

(D) $q(\theta) = (1 + \theta^{-\gamma})^{\gamma}$

(E) $q(\theta) = (1 - \theta^{-\gamma})^{-\frac{1}{\gamma}}$

(F) None of the above

$$q = \frac{(U^{-\gamma} + V^{-\gamma})^{-\frac{1}{\gamma}}}{(V^{-\gamma})^{-\frac{1}{\gamma}}}$$

$$= \left[\frac{U^{-\gamma} + V^{-\gamma}}{V^{-\gamma}}\right]^{-\frac{1}{\gamma}}$$

$$= \left[\frac{U^{-\gamma}}{V^{-\gamma}} + 1\right]^{-\frac{1}{\gamma}}$$

$$= \left[\left(\frac{1}{\theta}\right)^{-\gamma} + 1\right]^{-\frac{1}{\gamma}}$$

$$= \left[1 + \theta^{\gamma}\right]^{-\frac{1}{\gamma}}$$